Redefining xG

Integrating Advanced Features for Improved Predictions

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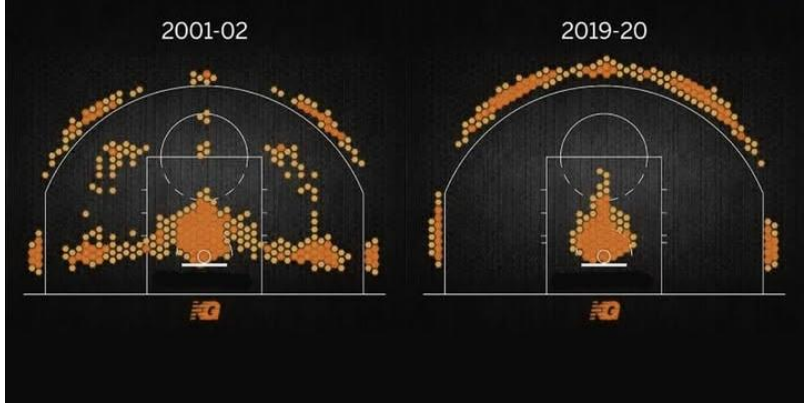
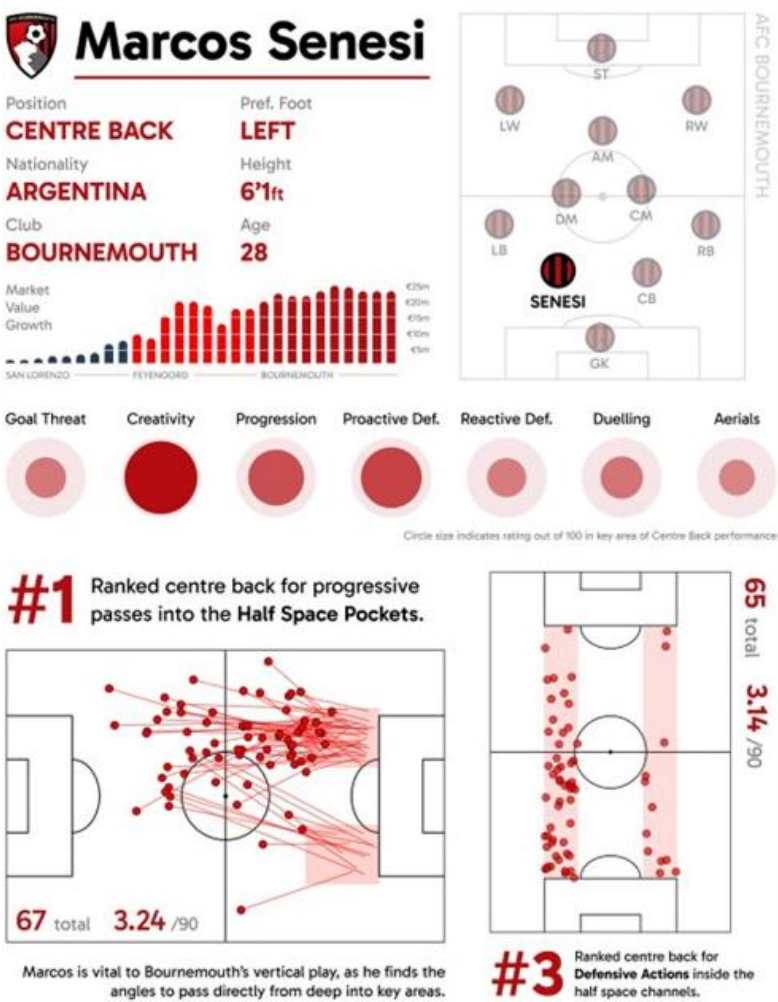
01

Introduction



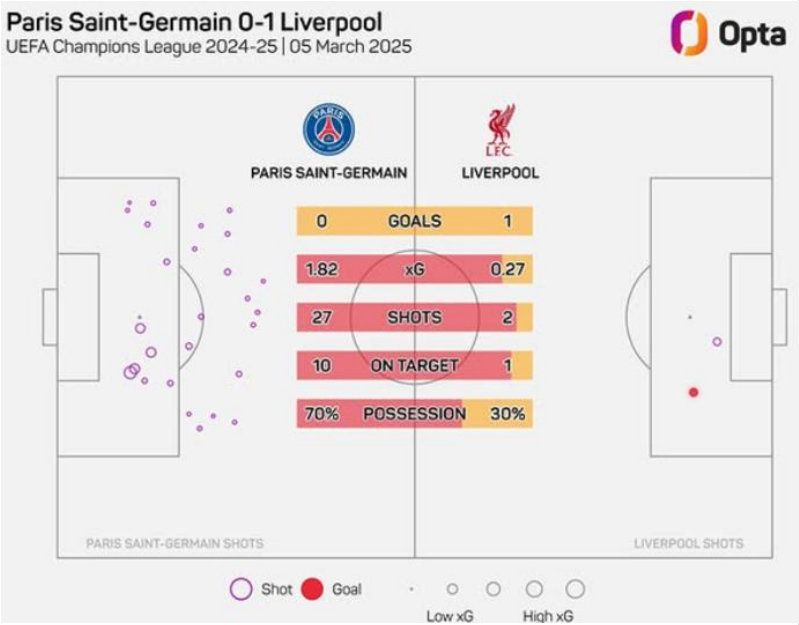


In these last years, several sports have been influenced by **Data Analysis**, to the point in which, in some of them, **Istudying data have revolutionized playstyles and market strategies**, allowing teams to **allocate money and play in a smart way**.

In Football, Data Analysis brought to the introduction of several **advanced metrics**, which can bring a **competitive advantage** on other teams, allowing to **analyze performance deeper,** and to **find hidden gems** at a low price.

Expected Goal

- ***Expected Goal*** (or *xG*) is a stat that defines a **probability for a certain shot to end up in a goal**
- It can be seen as a **conditional probability,** which depends on the **shot context**
- This index is only a **prediction,** an **estimate**
- *xG* allows to determine how many goals the teams *should have scored,* **going further than the simple result of a match**, which could often depend on **randomness**



Problem Definition

- xG are computed through **statistical models,** based on datasets which include hundreds of thousands of shots, with many features
- **Current models fail to capture the context around the shot**, producing an estimate **for the average player**
    - The opposition's positioning could be crucial
    - Factors related to the players' skills could add useful information
- *Integrated player-related and team-related data could allow to obtain more accurate predictions?*

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Implementation

Used Data

- Data have been gathered from a single season, for the 5 top European leagues
  - Serie A (Italy), Premier League (England), Bundesliga (Germany), La Liga (Spain), Ligue 1 (France)
- Each league-specific dataset includes **between 8k and 10k shots**
- Data have been collected from different sources (*StatsBomb, FotMob, SoFIFA, ClubElo*), also resolving some conflicts between information of different platforms

Dataset Types

Baseline

Used for simple predictions, only includes the shot's **distance** and

Intermediate

Includes many other factors, both **contextual** (situation, body part) and **player-related** and **team-related**

Full (Improved)

Includes features describing the **opposition's positioning,** later quantifying the **distance and number** of opposite players around the shooter (in the *Improved* version)

Dataset Types

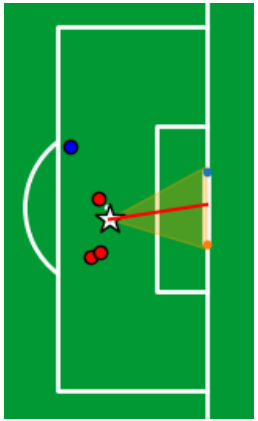
Distance	*Angle*
18	19

Minute	*Goal Difference*	*Rating*	*Elo Team*	*Keeper Rating*	*Elo Opponent*	*isHome*	*Position*	*Situation*	*Body Part*
26	3	70	1636	76	1607	True	Forward	Counter	Strong Foot

One on One	*Open Goal*	*Pressure*
False	False	True

Opponents in Cone	*Minimum Distance*
1	1.5

Shot Frame

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Algorithms and Approaches

- Many *preprocessing* operations have been used, like *normalization* and *One-Hot Encoding*
- Different algorithms have been tested, using both *neural networks* and *ensemble methods*
    - *Logistic Regression*, *XGBoost*, *Random Forest*, *AdaBoost*, *LightGBM*, *Multi-Layer Perceptron*
- A *parameter tuning* has been used on every model, to obtain the best possible configuration, through a GridSearch, and a *Cross Validation* has also been implemented, to test the model's stability, splitting data in folds based on the matches, achieviing great results

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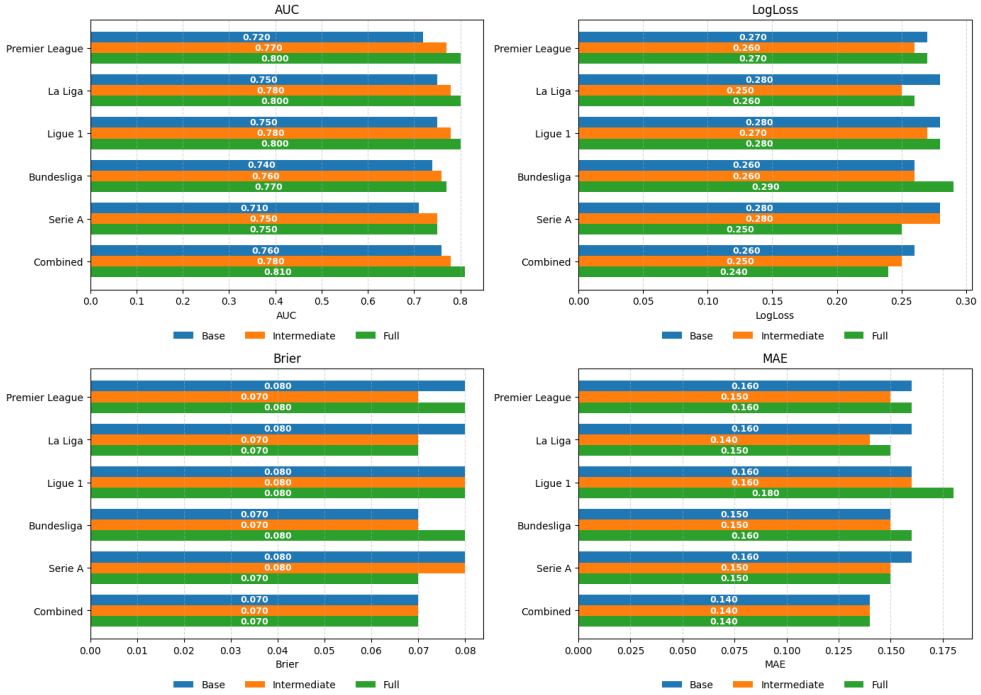
Experiments

Performance By Dataset

Every league's dataset has been used to train a specific model, and *within-league* metrics have been tested. **The *full* dataset does not always bring improvements.** (*XGBoost* has been used for all tests)



Performance By Dataset

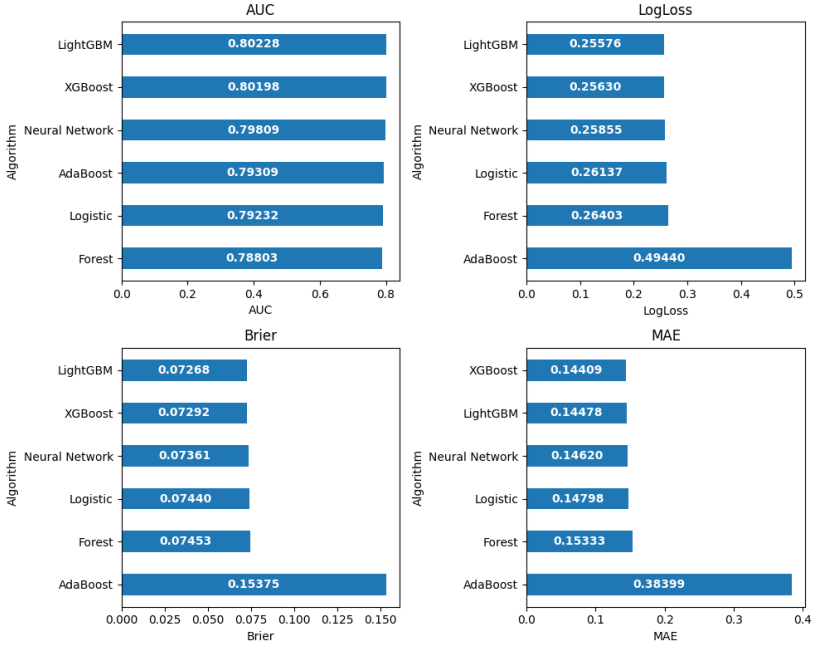
	ENG	ESP	FRA	GER	ITA	Combined
1	Distance	Distance	Distance	Distance	Distance	Distance
2	StrongFoot	Shooter	Head	Head	Head	Head
3	Shooter	Angle	OpenGoal	Angle	Angle	OpenGoal
4	EloOpp.	Head	Angle	Corner	EloTeam	Shooter
5	Angle	OpenGoal	Corner	StrongFoot	FreeKick	Angle
6	Head	EloOpp.	Shooter	EloTeam	StrongFoot	StrongFoot
7	OpenGoal	Keeper	Pressure	Pressure	EloOpp.	Corner
8	Corner	Pressure	Keeper	Shooter	Shooter	FreeKick
9	Minute	StrongFoot	EloOpp.	OpenGoal	Corner	Pressure
10	Pressure	Counter	EloTeam	Counter	Pressure	Counter

Performance By Dataset

	ENG	ESP	FRA	GER	ITA	Combined
1	Distance	Distance	Distance	Distance	Distance	Distance
2	MinDist	OppInCone	OppInCone	OppInCone	OppInCone	OppInCone
3	OppInCone	MinDist	Head	MinDist	MinDist	MinDist
4	Angle	Shooter	MinDist	Head	Angle	Angle
5	EloOpp.	Angle	Angle	Angle	Head	Head
6	StrongFoot	Head	Keeper	StrongFoot	Shooter	StrongFoot
7	Head	EloOpp.	EloTeam	EloTeam	StrongFoot	Shooter
8	Shooter	StrongFoot	Shooter	Counter	EloTeam	keeper
9	Minute	Keeper	StrongFoot	EloOpp.	Keeper	EloTeam
10	EloTeam	Minute	Minute	Forward	EloOpp.	Corner

Performance By Algorithm

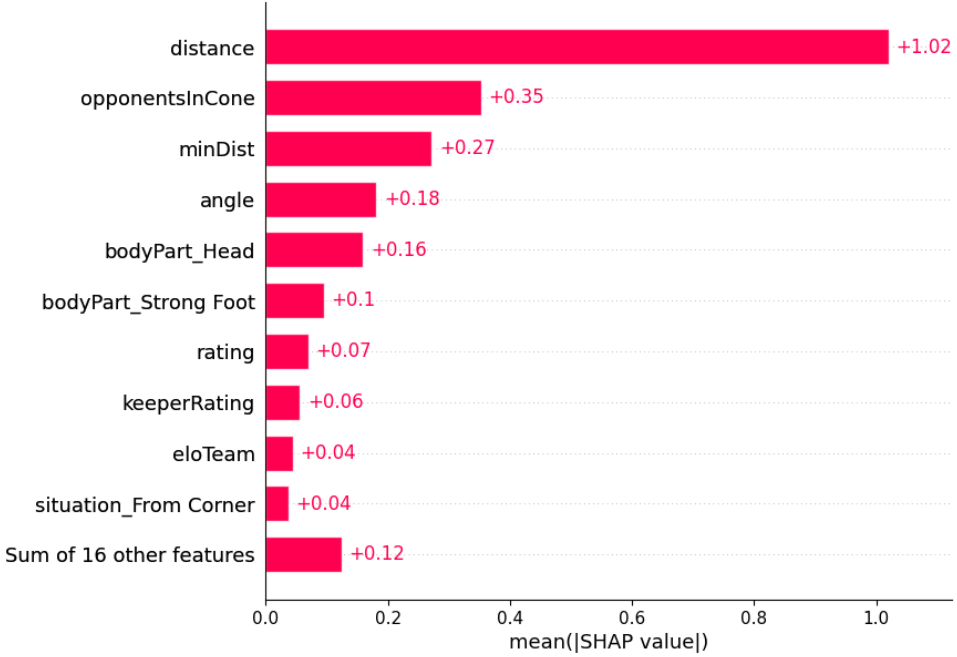
For each algorithm, a model has been trained on the TOP5 metrics, and the *within-league* performances have been analyzed.



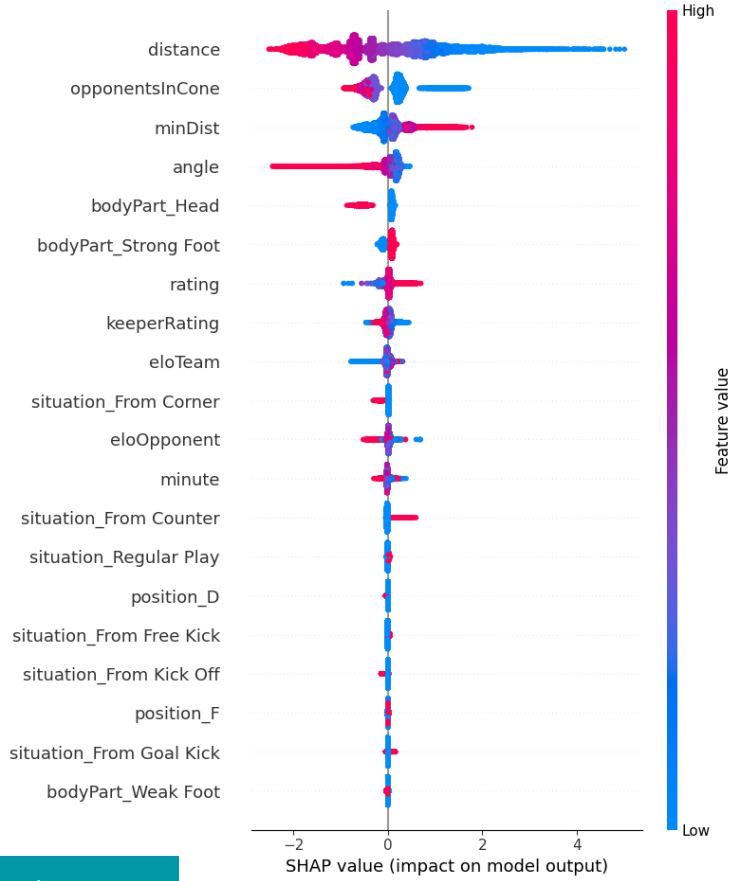
Performance By Algorithm

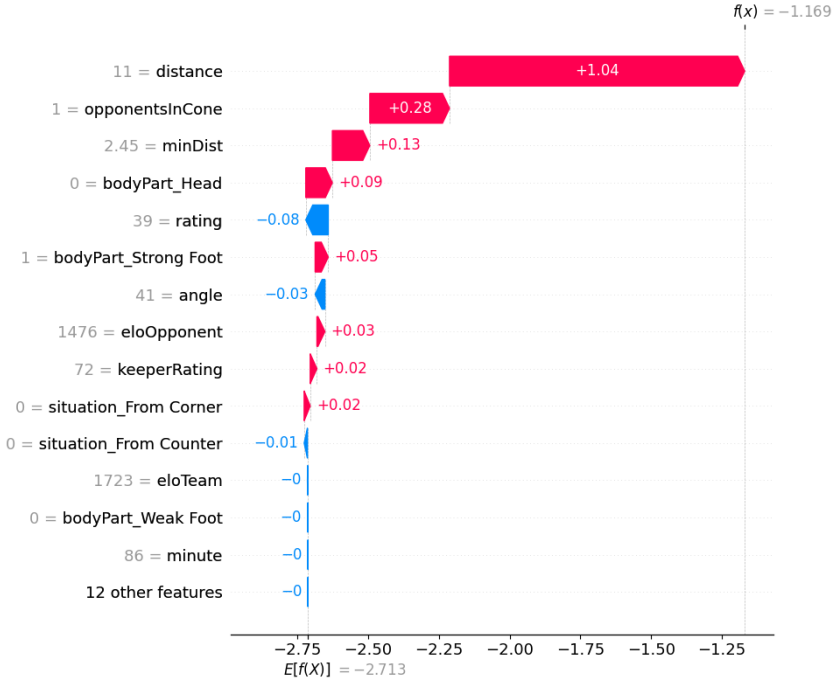
For both algorithms, a model for each league (excluding Serie A) has been trained, observing the *cross-league* metrics and the produced xG values. Despite LightGBM is better in terms of metrics with bigger datasets, **XGBoost proves to be more stable with smaller datasets.**

Model Interpretation



Model Interpretation





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Results

Comparing Models and Competitors

Predictions on Single Shot

Model	*MSE* ↓	*MAE* ↓	R^2 ↑
StatsBomb	**0.06553**	**0.13**	**0.23802**
Full (XGB)	0.06707	0.13739	0.22013
Full (LGBM)	0.06711	0.13739	0.21973
Intermediate	0.06966	0.14462	0.19003

Aggregate Predictions for Single Player

Model	*MSE* ↓	*MAE* ↓	R^2 ↑
StatsBomb	0.8588	1.69697	0.85138
Full (XGB)	0.85497	1.73872	0.84772
Full (LGBM)	**0.844**	**1.66607**	**0.85409**
Intermediate	0.89163	1.82071	0.84054

The proposed models achieve **great results, not better than the competitor's, but close,** despite the way smaller amount of data and features used

Some models perform better than competitors

Conclusions

- Adding player-related and team-related information is not enough to achieve satisfying results, as **these features influence models, but with a smaller impact compared to other factors**
    - The contextual variables (regarding the **ball and opposition location**) are still crucial to determine probabilities
- Despite this, it has been proven how **great models could be built also with less data,** by using effective techniques and **providing significant**